

Regenerative Dynamic Soaring Trajectory

Augmentation over Flat Terrains

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A gliding technique, known as dynamic soaring (DS), replicates the flight pattern of the wandering albatross bird to enable energy neutral, repeatable flight trajectories. This study investigates the potential for DS to act as a basis for UAV battery power regeneration by means of a windmilling propeller placed on the nose of the aircraft. In order to give an indication of the type of atmospheric and environmental conditions necessary to perform regenerative dynamic soaring (RDS), trajectories were optimized for the DT-18 drone. The optimal flight paths for varying surface conditions and periodic constraints are presented and compared to a base, energy neutral DS case. The findings suggest that by slightly altering the DS flight pattern, RDS is possible with significant battery recharge levels being reached for the DT-18 depending on wind strength and surface conditions.

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Nomenclature

C_L	= lift coefficient	
D	= drag force, N	
D_{gen}	= regenerative drag force, N	
E_{bat}	= battery energy charge, J	
F_x	= force in the x direction, N	
F_y	= force in the y direction, N	
F_z	= force in the z direction, N	
K	= induced drag factor	
L	= lift force, N	
m	= mass, kg	
N_t	= number of time steps	
P_{aero}	= aerodynamic power, W	
P_{gen}	= regenerative power, W	
$P_{S\&L}$	= minimum power expenditure for straight and level flight, W	
S	= wing surface area, m ²	
t	= time, s	
t_{cycle}	= time of a cycle, s	
T	= thrust force, N	
U_{ref}	= reference wind speed, m/s	
V_a	= airspeed, m/s	
V_i	= ground speed, m/s	
V_w	= wind speed, m/s	
x	= northern displacement, m	
y	= eastern displacement, m	
z	= negative altitude, m	
z_0	= surface roughness length, m	
z_r	= reference wind height, m	
η_{gen}	= total wind-prop chain efficiency	
γ	= flight path angle, rad	2
ϕ	= bank angle, rad	
ψ	= heading angle, rad	
ρ	= density, kg/m ³	

I. Introduction and Objectives

Drone technology is rapidly advancing in many domains, however, technological challenges remain when it comes to increasing the range and endurance of UAVs. Both range and endurance are largely dependant on the power source available on-board the aircraft, which is often restricted by payload mass and size requirements.

Solutions in engineering are often taken directly from nature, and in the case of aerospace engineering, from birds. Albatrosses are a genus of bird which spend extended periods of time foraging for food over the open ocean. The Wandering Albatross (*Diomedea Exulans*) has been observed flying thousands of kilometres when foraging for food, with daily ranges exceeding 900km and some even being recorded flying a complete circle of latitude around the Southern Ocean [1]. Without consuming enough food energy to maintain flapping wing flight over such large distances, it became apparent that the albatrosses were using a gliding method to traverse their oceanic terrain. This flying technique was labelled dynamic soaring (DS). Sachs et. al. presented the first experimental results on the small-scale movements that make up DS [2], and showed that it is composed of four progressive flight phases. The first phase is a climb into the wind. The second is an upper turn from a windward flight direction to a leeward flight direction. The third is a descent with the wind. The final phase is a lower turn from leeward back to windward flight. These maneuvers allow for an energy neutral, repeatable flight cycle, and are enabled by the strong vertical wind gradients present in the boundary layer over the ocean. Sachs then developed a mathematical method for optimizing dynamic soaring trajectories [3]. Using a logarithmic wind profile to simulate the wind field in the boundary layer over the ocean, he converged on a DS flight path for the wandering albatross given the objective of minimizing the wind strength required to complete an energy-neutral flight cycle. The results produced a flight pattern similar to those recorded of the albatrosses they monitored over the ocean.

The same concept was then applied to UAVs. A study by Deittert et. al. optimized DS trajectories for a small UAV while minimizing the reference wind speed, and enforcing a number of displacement constraints [4], including one in a closed-loop. Extended endurance while flying closed-loop trajectories poses benefits for certain mission objectives, such as surveillance or search

and rescue operations. The solved closed-loop trajectory resembled an eight-shape when observed from above. This pattern was shown to produce the same four DS maneuvers repeated twice within one cycle.

Taking a step further than energy-neutral flight would then be to create a flight cycle which experiences a net energy *extraction*. Glauert (1926) first introduced the concept of regenerative soaring, suggesting that the use of a windmill on an airplane has potential [5]. 72 years later, MacCready [6] discussed the possibility of applying the regenerative braking technologies used in battery powered cars to a sailplane propeller acting as a windmill during flight. More recently, Barnes [7] [8] gave a detailed analysis of the performance of a dual-mode, windmilling propeller that uses regenerative braking technology, created a concept design for a UAV capable of regenerative soaring, and discussed the promise of DS based regenerative soaring. The first regenerative dynamic soaring (RDS) trajectory optimization study was, however, done by Bonnin et. al. at ISAE-SUPAERO [9]. To account for the additional drag produced by a windmilling propeller, an extra force called the regenerative drag force (D_{gen}) was introduced and simulated as a control variable. They chose the leeward side of hill as their terrain and optimized a closed-loop trajectory to be performed in the lowest possible wind strength.

While the particular wind profile which develops on the leeward side of a hill presents advantages for RDS, the scope of potential UAV mission scenarios can be broadened when RDS over a flat surface is considered. Thus the primary objective of this study is to establish if RDS is possible over flat terrains of varying topography. Both open-loop (net displacement) and closed-loop (fixed) trajectories were solved for to have a greater understanding on the practicality of RDS.

In order to achieve this, a detailed methodology is outlined that includes the aircraft model, the wind model and types of terrains studied, the kinematics, kinetics and powers associated with the aircraft system, and the optimization setup. Following this, a validation of the solver is presented, a comparison between open-loop and closed-loop DS is made, and the results and observations for the RDS trajectories over varying terrains are given.

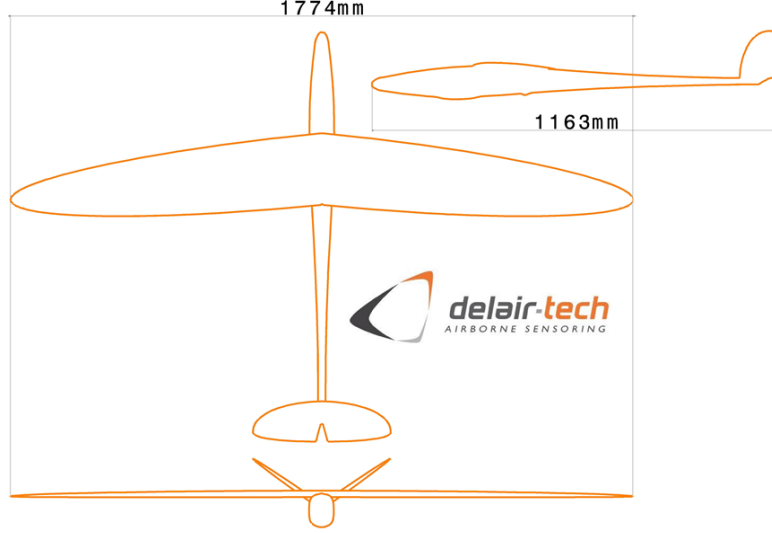


Fig. 1 Delair-Tech DT-18 UAV (reproduced from Bonnin et. al. [9]).

II. Methodology

A. Aircraft Model

The wandering albatross has the largest aspect ratio (16.8) of any bird, giving it an appearance and physical shape comparable to a common glider. A DT-18 was selected for this study which is a small UAV developed by Delair-Tech in Toulouse, France. It has a slender body and relatively large aspect ratio (see Figure 1 and Table 1). Furthermore, the DS performance of the DT-18 has already been explored throughout Bonnin's Ph.D. dissertation [10], and used for the research into regenerative dynamic soaring on the leeward side of a hill by Bonnin et. al. [9].

A point mass model was used to represent the aircraft in order to reduce the complexity of the model and to permit more of a generalization of the results. All motion, forces, and moments associated with the aircraft are thus assumed to act on the same point, namely the centre of gravity, and bank angle is the only rotation about this point which is taken into account. It is further assumed that the aircraft has a propeller mounted on its nose, aligned with the airspeed vector, which is capable of switching into windmilling mode in order to generate electricity.

Table 1 DT-18 parameters (reproduced from Bonnin [10])

Parameter	Value
Mass (kg)	1.7
Payload (kg)	2.0
Wing span (m)	1.8
Wing area (m^2)	0.248
Wing loading (N/m^2)	146.4
Aspect ratio	13.1
C_{D_0}	0.0259
$C_{L_{max}}$	1.195
$(L/D)_{max}$	15
Min. power consumption for S&L flight (W)	42.4

B. Wind Model and Terrains

DS is reliant on vertical wind shear (wind gradient), which develops in the boundary layer over any surface. Various approximate models have been applied to wind profiles over flat terrains (with varying topographies) including linear, exponential and logarithmic relationships, however, Bonnin [10], Barnes [7], [8], and Sachs [3] all opted to use a logarithmic profiles due to the altitudes reached by albatross flying DS trajectories and the wind strengths required to effectively perform DS. It should be noted that the logarithmic wind profile approximation is unrealistic as wind turbulence will undoubtedly affect the overall flight dynamics of an aircraft, and may reduce the time-averaged performance of the trajectories. For the purposes of this exploratory study, however, the logarithmic profile has been deemed applicable.

$$V_w = U_{ref} \cdot \frac{\ln(-z/z_0)}{\ln(z_r/z_0)} \quad (1)$$

Equation 1 describes the relationship between wind strength and altitude, and reveals the necessity for a reference wind height and corresponding wind speed when determining the wind strength. A reference wind height of $10m$ was chosen for this study, the same chosen by Sachs for his optimization [3]. Equation 1 also exposes the reliance of the logarithmic scale to the roughness length at the surface. Surface roughness lengths are dependent on the type of terrain protruding from

Table 2 Surface conditions [11]

Terrain	Surface Roughness (m)
Open flat plain / low grass	0.03
Low crops	0.10
Subtropical savannah / low scattered trees	0.20
Suburban	0.30
City: Commercial	0.56
City: Residential	0.73

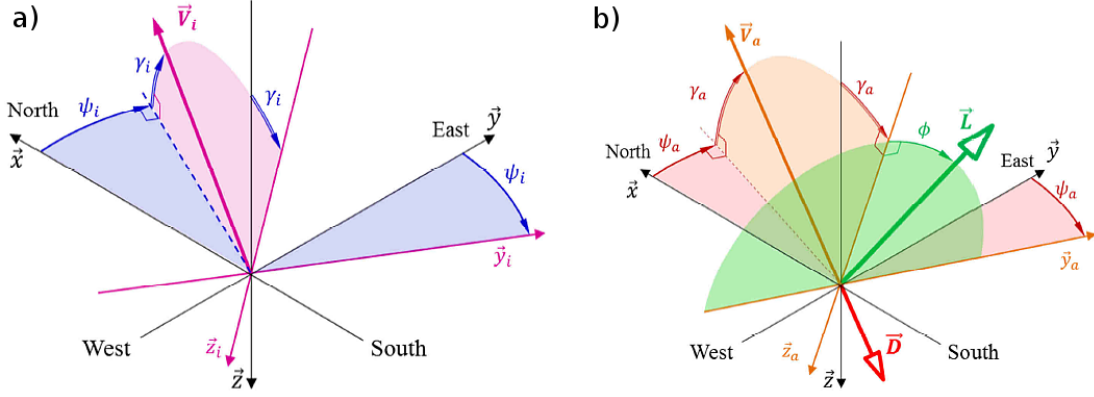


Fig. 2 a) Ground speed coordinate system. b) Airspeed coordinate system. (Reproduced from Bonnin [10])

the ground. Table 2 gives examples of several types of terrains and their corresponding roughness lengths, derived from Choi [11].

C. Kinetics and Kinematics

Although a debate exists into whether DS should be analyzed via an inertial or wind-based frame of reference, the inertial reference frame was selected for this research as it seems more intuitive due to the necessity of Earth-based performance results for real world applications. The Earth-based (inertial) reference frame corresponds to a change in wind orientation being the root of DS energy gains, rather than a change in magnitude (Sachs et. al. argue that the principle energy gain is seen during the turn from upwind to downwind [2]).

Figures 2a and b portray the relationship between the coordinate system, the ground velocity

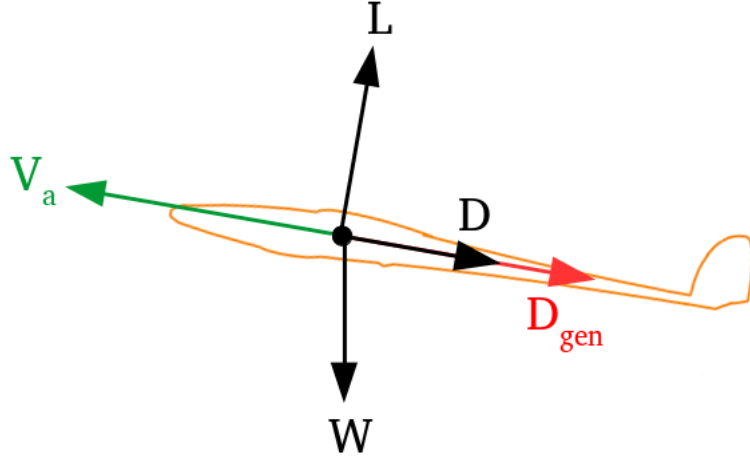


Fig. 3 Forces acting on the aircraft.

vector, the airspeed vector, and the aerodynamic forces [10]. The coordinate system aligns $\vec{x}$ with North, $\vec{y}$ with East, and $-\vec{z}$ with positive altitude. The relationship between the airspeed and ground speed vectors is established in Equations 2.

The inertial state variables $(V_i, \psi_i, \gamma_i, x, y, z)$ describe the dynamic state of the aircraft and are used to define the evolution of the flight trajectory at any given moment via the equations of motion (Equations 7 to 12). The state variables are complemented by the control variables $(C_L, \phi, \text{and } D_{gen})$ which govern the state variables during the trajectory's progression, and are considered manipulable by the pilot or control system of the aircraft. D_{gen} designates the additional drag force acting on the vehicle when the propeller is functioning in windmilling mode. As can be seen in Figure 3, D_{gen} acts in the same sense and direction as the aerodynamic drag force.

$$V_a = \sqrt{V_i^2 - 2V_iV_w \cos \psi_i \cos \gamma_i + V_w^2} \quad (2)$$

Equation 2 describes the relationship between the airspeed and ground speed in the inertial reference frame, as well as introducing the relationship to the wind strength.

$$\vec{F}_{ext} = F_x \vec{x} + F_y \vec{y} + F_z \vec{z} \quad (3)$$

$$F_x = -L(\sin \phi \sin \psi_a + \cos \phi \cos \psi_a \sin \gamma_a) - (D + D_{gen})(\cos \gamma_a \cos \psi_a) \quad (4)$$

$$F_y = L(\sin \phi \cos \psi_a - \cos \phi \sin \psi_a \sin \gamma_a) - (D + D_{gen})(\cos \gamma_a \sin \psi_a) \quad (5)$$

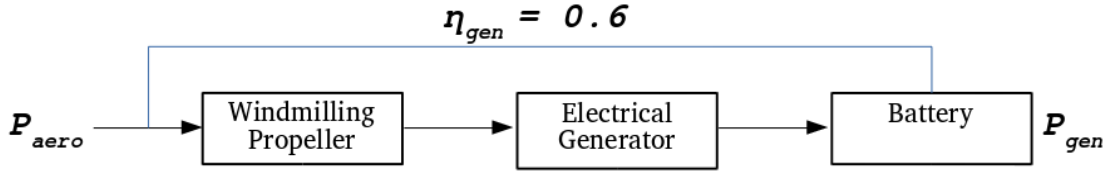


Fig. 4 Regenerative power chain.

$$F_z = -L \cos \phi \cos \gamma_a + (D + D_{gen})(\sin \gamma_a) + mg \quad (6)$$

Equations 3 to 6 characterize the external forces acting on the aircraft and highlight how D_{gen} is summed with the aerodynamic drag force when computing the x, y, and z positions.

$$m\dot{V}_i = F_x \cos \gamma_i \cos \psi_i + F_y \cos \gamma_i \sin \psi_i - F_z \sin \gamma_i \quad (7)$$

$$mV_i \cos \gamma_i \dot{\psi}_i = -F_x \sin \psi_i + F_y \cos \psi_i \quad (8)$$

$$-mV_i \dot{\gamma}_i = F_x \sin \gamma_i \cos \psi_i + F_y \sin \gamma_i \sin \psi_i + F_z \cos \gamma_i \quad (9)$$

$$\dot{x} = V_i \cos \gamma_i \cos \psi_i \quad (10)$$

$$\dot{y} = V_i \cos \gamma_i \sin \psi_i \quad (11)$$

$$\dot{z} = -V_i \sin \gamma_i \quad (12)$$

The equations of motion describe the evolution of the system during flight, and in particular for this study can be analyzed to understand how D_{gen} affects the motion of the vehicle.

D. Power Chain

Figure 4 exhibits the power chain of the power regeneration system. The aerodynamic power imparted on the windmilling propeller during flight is equal to the product of the negative of D_{gen} and the airspeed vectors. This is not equal to the battery recharge rate as the various power conversions required incur efficiency losses. Thus, for simplicity, the regeneration efficiency of this

entire process is taken as a constant of 0.6, taken as the maximum power conversion efficiency of a windmill, defined by the Betz limit [13].

$$P_{aero} = -\vec{D}_{gen} \cdot \vec{V}_a \quad (13)$$

$$P_{gen} = \dot{E}_{bat} = (\vec{D}_{gen} \cdot \vec{V}_a) \eta_{gen} \quad (14)$$

$$P_{net} = \frac{\Delta E_{bat}}{t_{cycle}} \quad (15)$$

The instantaneous battery recharge rate is also labeled as the regeneration power. The net income power (P_{net}) is a parameter which specifies the average battery recharge rate during one RDS cycle and is used to calculate the amount of power which can be harvested during the flight.

E. Optimization Setup

The trajectories were optimized using The Sparse Nonlinear OPTimizer (SNOPT) solver, offered by the NEOS server. As SNOPT takes first order-derivatives with a specific optimization structure as input, the modelling tool, A Mathematical Programming Language (AMPL), was used to convert the equations of motion, constraints, and objective function into the appropriate format. A MATLAB script interfaced the input and output data from the coupled SNOPT-AMPL package. SNOPT finds only the local optimal solution which must be considered when analyzing and interpreting the results. This setup was replicated from Bonnin's dissertation [10].

The constraints are input to the solver as a series of bounds (Table 3) and periodic conditions which were designed to ensure that the trajectories were realistic and physically feasible. Periodicity is applied as a set of final condition constraints. For the open-loop case, the variables z , γ , ψ , ϕ , C_l , V_i and D_{gen} are all constrained such that the final condition must equal the initial condition. The same is true for the closed-loop case, however, with the addition of the x and y variables. The average battery charge rate was imposed as final condition constraint.

An objective function defines a variable or parameter which is to be maximized or minimized by the solver. The solver is able to manipulate all input variables, while complying with the subjected constraints, to output the optimal trajectory that achieves the objective function. As DS is thought

Table 3 Trajectory optimization constraints

Constraints	Values
No stall	$0 \leq C_l(t) \leq C_{l_{max}}$
Maximum load factor	$\frac{1}{2} * \rho * S * C_l(t) * V_a(t)^2 \leq 3 * m * g$
Flight path angle bounds	$-75^\circ \leq \gamma(t) \leq 75^\circ$
Heading angle bounds (Open-loop)	$0^\circ \leq \psi(t) \leq 180^\circ$
Heading angle bounds (Closed-loop)	$-90^\circ \leq \psi(t) \leq 360^\circ$
Bank angle bounds	$-85^\circ \leq \phi(t) \leq 85^\circ$
No thrust	$T(t) = 0$
Maximum power generation (RDS)	$D_{gen}(t) * V_a(t) * \eta_{gen} \leq 300W$
Ground clearance	$1.5m \leq z(t)$

Table 4 Comparison with Sachs' albatross trajectory optimization

Variable	Sachs' Results [3]	Bonnin's Results [10]	% Difference
$U_{ref} (m/s)$	8.6	8.59	0.1
$t_{cycle} (s)$	7.1	7.0102	1.3
$\Delta z_{max} (m)$	20.5	0.20.09	2.0
$\Delta y_{max} (m)$	66.5	65.78	1.1
$\Delta x_{max} (m)$	66.2	64.46	2.7

to be highly dependent on the surrounding wind field, the decision was taken to set minimizing the reference wind speed at the reference height of 10m as the objective function throughout this study. Determining the wind strength required to perform RDS helps to deepen knowledge on the type of environmental conditions necessary for RDS to be possible. Moreover, it parallels the objective functions of Sachs and Deittert et. al. [3], [4].

III. Dynamic Soaring Validation and Trajectory Comparison

To validate the methodology setup and reliability of the solver, Bonnin [10] ran a DS case using the wandering albatross as the vehicle model so that a comparison with Sachs' solution could be made. The optimization used an identical objective function, wind profile, reference wind height,

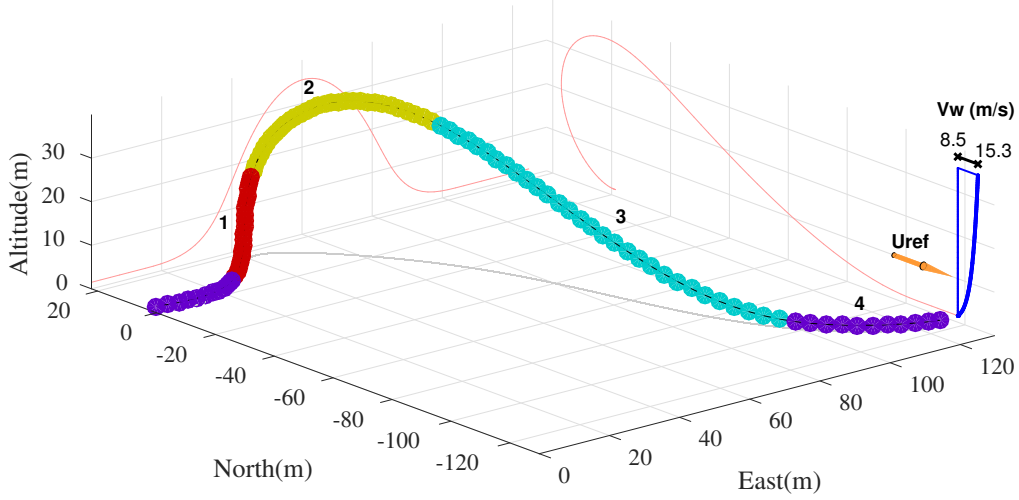


Fig. 5 DT-18 open-loop dynamic soaring trajectory.

surface roughness (3cm), frame of reference, and albatross model. The resulting trajectory showed a strong resemblance to that found by Sachs [3]. The objective function returned a value with only 0.1% variation with Sachs, while the maximum displacement differences, and overall cycle time were all within 3%. The optimization setup and methodology are, therefore, considered valid based on previous literature, and to represent trajectories that hold physical meaning. It is assumed that the same methodology applied to the DT-18 (different vehicle parameters) would remain valid.

Once the vehicle model was replaced with the DT-18 and the simulation was run, the drone replicated the same DS pattern flown by the albatross model. The four DS flight phases are highlighted in Figure 5 with red corresponding to ascent into the wind, yellow to the turn from windward direction to leeward direction, blue to descent with the wind, and purple to the turn from leeward direction to windward direction.

Additional periodic constraints were then input into the solver and a closed-loop trajectory was computed. Figure 6 gives the optimal trajectory for the DT-18 when flying in a closed-loop. In this case, the four DS movements are repeated twice in one cycle. It also represents a similar eight-shaped pattern to Deittert et. al. when viewed from above [4].

Table 5 highlights some of the discrepancies between open and closed-loop trajectories. The

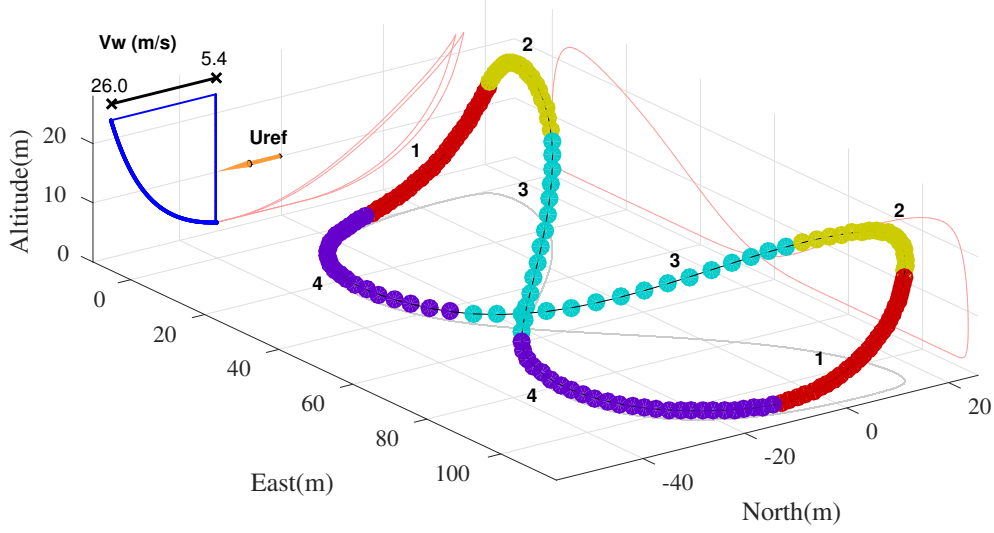


Fig. 6 DT-18 closed-loop dynamic soaring trajectory.

Table 5 Dynamic soaring trajectory comparison

Parameter	Open-Loop	Closed-Loop
$U_{ref} \text{ (m/s)}$	12.55	19.66
$t_{cycle} \text{ (s)}$	9.96	18.96
$\Delta z_{max} \text{ (m)}$	33.93	21.67
$\Delta y_{max} \text{ (m)}$	124.72	112.33
$\Delta x_{max} \text{ (m)}$	117.58	47.65

reference wind speed required to complete the closed-loop trajectory is significantly higher than that of the open-loop trajectory. The effect of the reference wind speed on the overall wind profile is projected in blue in Figures 5 and 6 where the values for the minimum wind speed (at lowest altitude) and values for maximum wind speed (at highest wind speed) are expressed on a scale at the top of the profile (in black). A much larger disparity is present for the higher reference wind speed of the closed-loop case. The closed-loop trajectory reaches a lower maximum altitude than the open-loop trajectory, however, the closed-loop case is still able to turn leeward at a higher wind speed due to the difference in wind profiles. The time to complete the closed-loop trajectory

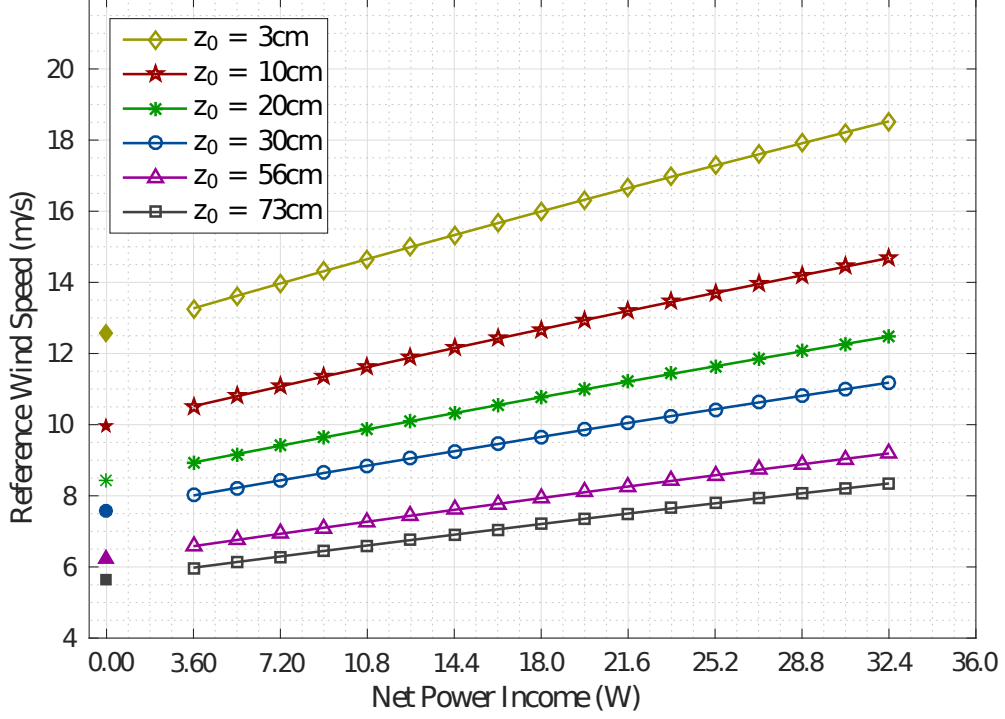


Fig. 7 Open-loop net power income vs. reference wind speed and surface roughness length.

is almost exactly twice that of the open-loop. This is interesting since the same flight pattern is completed twice, the distances flown and the wind profile differs.

IV. Regenerative Dynamic Soaring: Open-Loop

After introducing the regenerative power into the optimization, a series of open-loop RDS trajectories were optimized. All terrains defined in Table 2 were considered for range of P_{net} values.

A. Wind Speed

An overview of the wind strengths required for the open-loop RDS flight cycles (Figure 7) reveals that as the roughness length is reduced, a higher reference wind speed is required to recharge the battery at the same average rate. Similarly, in order to regenerate a higher rate of energy for a given surface roughness, a higher reference wind speed is required. A linear trend between P_{net} and reference wind speed was found, even for the zero P_{net} , pure DS case (represented in Figure 7 as solid markers of the same style).

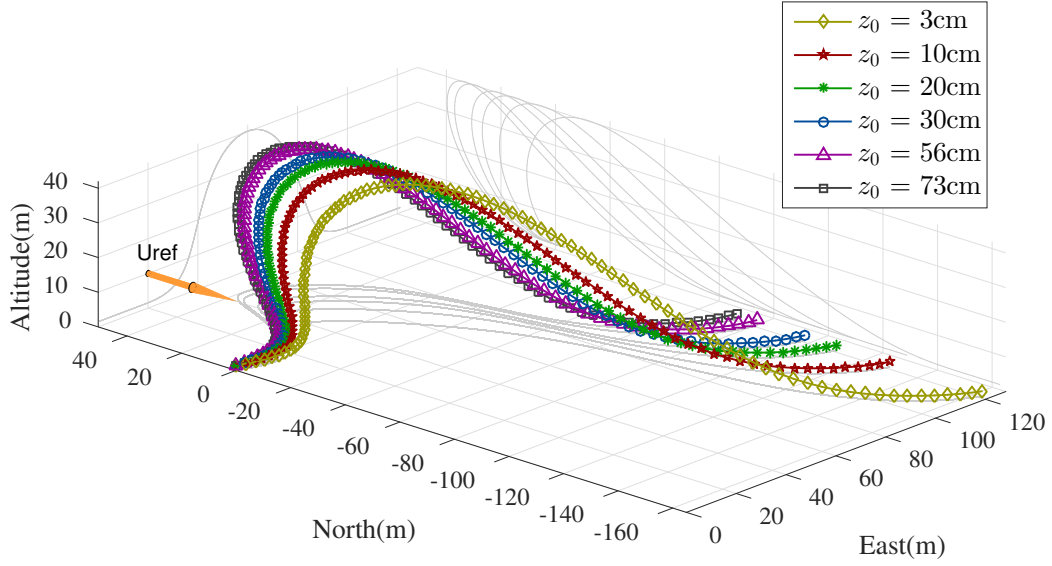


Fig. 8 Open-loop 3D flight paths for variable roughness lengths.

When comparing the wind speeds required for open-loop RDS with actual mean surface wind speeds, it can be seen that RDS is only feasible under limited circumstances and in certain locations. Yin [12] presented annual global surface wind speed data from 1506 weather stations recorded (mostly) at 10m in altitude. This data indicates that annual mean wind speeds, at 10m in height, generally remain below 6m/s, with the highest in this range ($>4m/s$) being found over the Great Central Lowland of North America, parts of the Western Siberian Lowland, and on the Arabian Peninsular. Yin did, however, find annual mean wind speeds of up to 16m/s in isolated, often landlocked locations.

B. Surface Roughness

Fixing the average battery recharge rate at 21.6W, a focus was placed on how the flight paths varied over increasingly rough terrains, as well as the matching evolution of variables with respect to time.

Figure 8 shows the open-loop RDS trajectories for each terrain type in the three-dimensional space. The orange arrow represents the height and orientation of the reference wind speed. The similarity between DS and RDS flight can immediately be observed; the same basic shape is retained

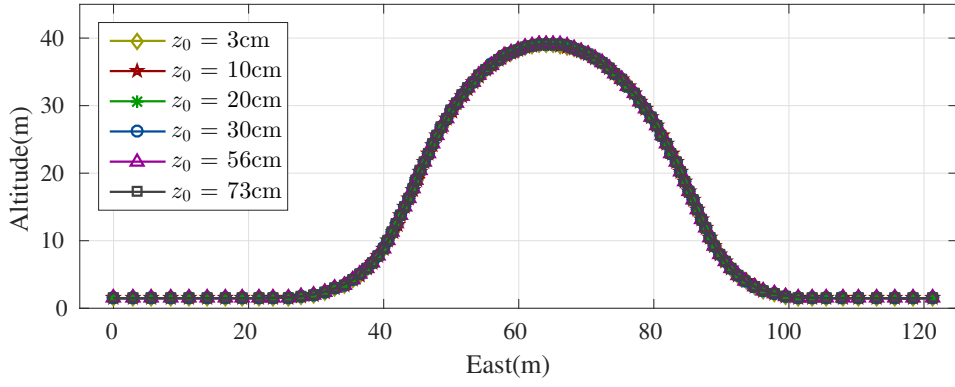


Fig. 9 Open-loop west-east flight paths for variable roughness lengths.

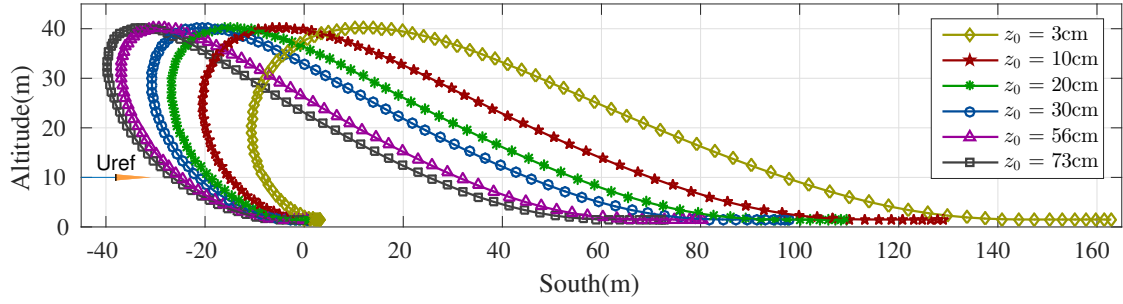


Fig. 10 Open-loop north-south flight paths for variable roughness lengths.

for all cases. Breaking the trajectories down into the two-dimensional space reveals the variation in flight dynamics. The west to east displacement (Figure 9) was identical for every surface terrain, displacing approximately 121m east. Furthermore, Figure 9 reveals the identical altitudes reached for each terrain, even though different wind strengths were present for the terrains at the same altitude due to the change in wind profile. The displacement from north to south (Figure 10), however, saw an increasing shift southward when the surface roughness length is reduced.

The airspeed was also found to remain constant over each terrain type, while the variation in ground speed varied conversely depending on whether flight was windward or leeward. The wind speed increased evenly with the variation in roughness length, based on the logarithmic nature of the wind profile (Figures 11a, b, and c).

The lift coefficient, bank angle, drag coefficient and regenerative power were each found to be constant over all terrains types. The maximum bank angle was reached during the upper turn for all of the cases. These results indicate that the ideal trajectory results from a specific combination

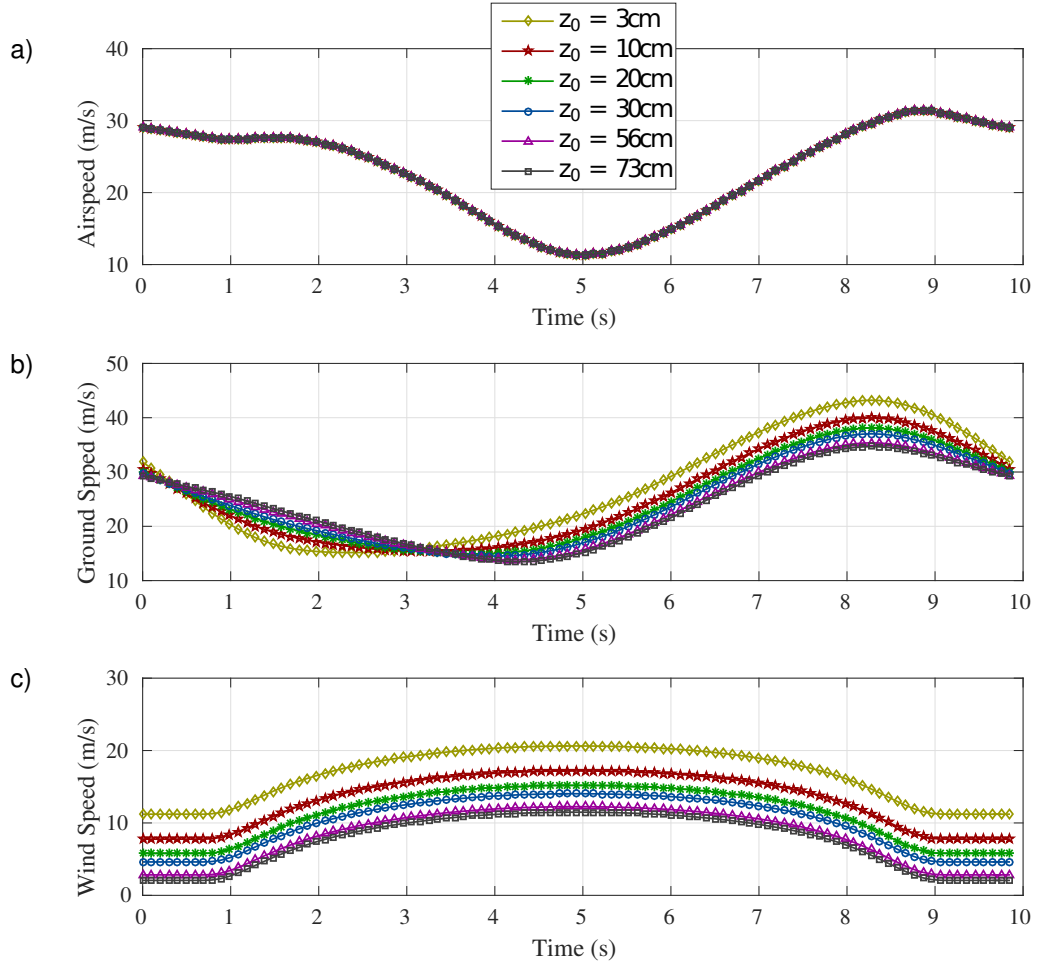


Fig. 11 Time versus: a) Airspeed. b) Ground speed. c) Wind speed.

of airspeed and D_{gen} in order to achieve the same average battery recharge rate. The airspeeds were maintained by manipulating the ground speed to compensate for the different strength wind fields which developed for each surface type. When referring back to Figure 10, it seems that the aircraft then traversed further south for the cycles with a lower surface roughness length to achieve the higher ground speeds. A higher acceleration is achieved by the extra tailwind and further displacement south within the same time period. Moreover, the aircraft spent less time flying against the wind as the surface roughness length was decreased due to the higher incoming wind speeds resulted the same airspeed. The displacement from west to east could also be attributed to the fact that the wind speed vector is at a perpendicular angle to the displacement, so has no effect on the movement in that plane, and wind strength is the changing factor between cases.

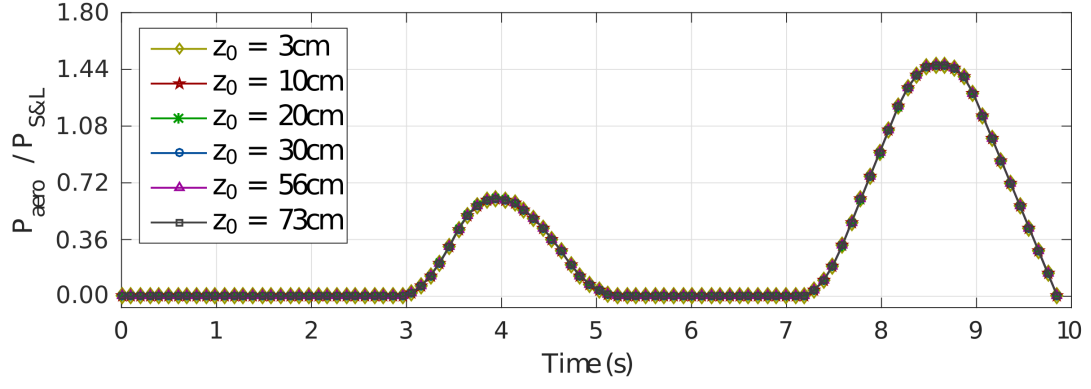


Fig. 12 Time versus aerodynamic power.

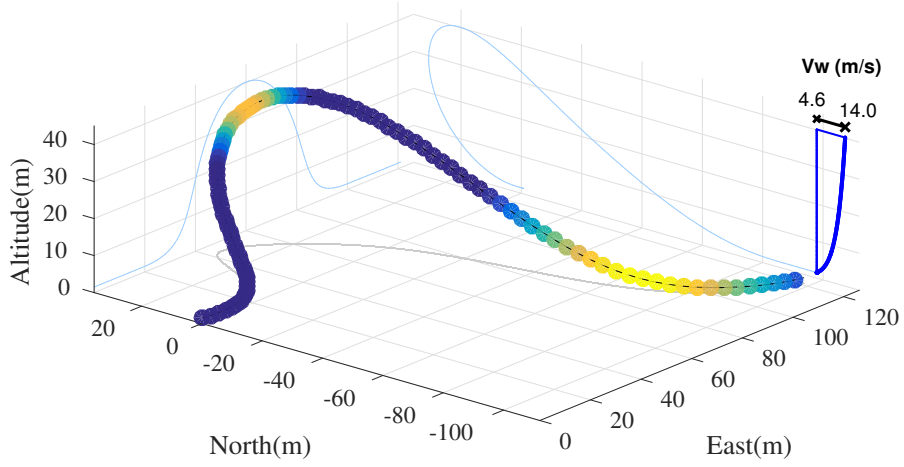


Fig. 13 Regenerative drag force along open-loop trajectory.

The variation of aerodynamic power with time has been non-dimensionalized with respect to the power expenditure experienced during powered straight and level (S&L) flight for maximum endurance, in Figure 12. The aerodynamic power imparted on the propeller reaches a level equal to the power required to sustain S&L flight ($42.4W$), and even surpasses it to approximately 1.44 times the amount of power during the descent with the wind (seen in Figure 13). This would suggest that for the same net displacement angle, the power that could be harvested from the wind would be lesser, but of the same magnitude, to the power the propeller would need to expend in order to maintain the aircraft in flight for maximum endurance, should an RDS trajectory be flown instead of flying S&L.

D_{gen} was applied at two points along the trajectory, as seen in yellow in Figure 13, the first

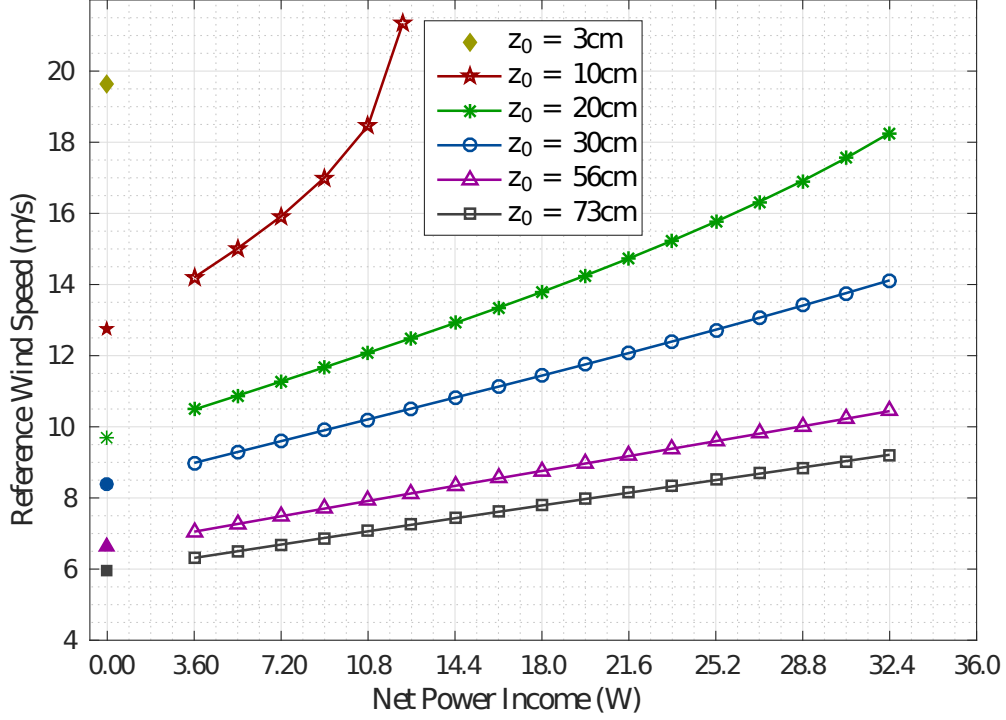


Fig. 14 Closed-loop net power income vs. reference wind speed and surface roughness.

on the turn from windward to leeward direction and the second during leeward descent. Although a similar amount of drag is applied in both instances, the difference in aerodynamic powers varies significantly, with a maximum difference of 0.72 times the power required for S&L flight. This is due to the difference in airspeed during the two maneuvers.

The overall time of each of the flight cycle for each terrain type remained constant (given a fixed P_{net}), even though the flight path varied for the differing topographies. For example, for the above case with a P_{net} of 21.6W, the final time was approximately 10s.

V. Regenerative Dynamic Soaring: Closed-Loop

After investigating the open-loop RDS cases, where the DT-18 was free to displace in the north-south and west-east directions, the displacements were constrained such that the x and y components have equal initial and final values, creating a closed-loop.

A. Wind Speed

As can be seen in Figure 14, RDS was not possible over low flat terrains ($z_0 = 3cm$). Even the DS closed-loop case required a comparatively high reference wind speed (approx. $20m/s$). Although RDS was shown to be possible over a terrain with a surface roughness length of $10cm$, it had a maximum P_{net} of approximately $12.6W$. Each of the other tested terrains were at least capable of regenerating up to an average battery recharge rate of $32.4W$, with all but the low scattered tree and low crop cases showing a similar linear relation to the open-loop RDS cases. As the power available for extraction in the system reached its limits, the wind strength required rapidly increased. The wind strength for each terrain type was higher than that required for open-loop RDS trajectories, as expected from the comparison of DS cases.

As discussed for the open-loop RDS cases, the wind speeds required for closed-loop RDS flight greatly limit the scenarios which would permit its application.

B. Surface Roughness

The effects of terrain on RDS performance for the closed-loop case were also tested with a fixed P_{net} of $21.6W$. This excluded trajectories over low flat terrain ($3cm$) and over low crops ($10cm$), however, this gives more reliable results as no battery recharge limits are being approached.

As with the open-loop cases, the closed-loop trajectories replicated the same basic pattern as the closed-loop DS trajectory. However, unlike for the open-loop RDS cases with varying roughness lengths, the crosswind displacement was not constant for flight over all terrains. Analyzing the two-dimensional projections in Figure 15 shows that the main variations between the trajectories were an increase in altitude, and a shift downwind, for the drone flight over higher roughness lengths. The trajectory was initialized at the origin on the north-south, east-west plane ($0m, 0m$), indicated by a black line in Figure 15.

Ground speeds remained fairly similar for all surface roughness lengths (Figure 16b). This could be achieved by the increase in altitude and displacement south for the higher surface roughness trajectories, allowing a longer distance for acceleration during descent with a tailwind. The airspeeds (Figure 16a) are, however, higher for the lower roughness lengths during the climbs, due to the higher

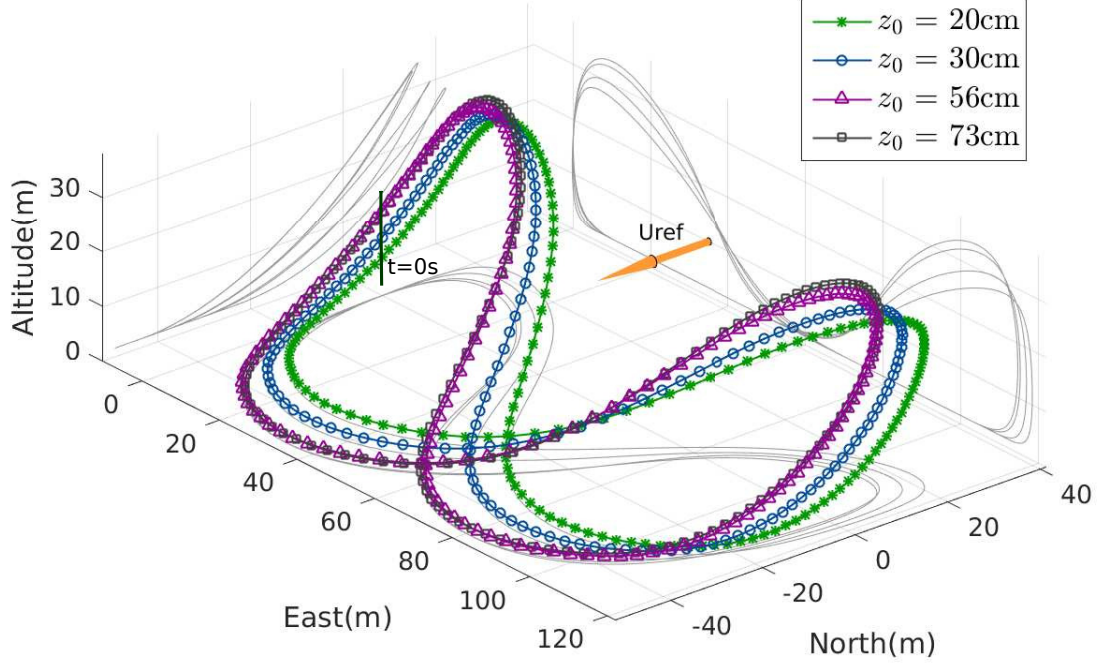


Fig. 15 Closed-loop 3D flight paths for variable roughness lengths.

incoming wind speeds. Once D_{gen} and P_{gen} were considered, it becomes apparent that the higher airspeeds for the lower surface roughness lengths were necessary to compensate for the lower levels of D_{gen} applied in order to maintain the same P_{net} value.

The bank angles, shown in Figure 17a, stayed fairly constant for all terrain types, as with the open-loop RDS cases. The maximum bank angle was reached during the upper turn in both loops, and same pattern was followed during both loops in the reverse direction. The evolution of the lift coefficient, in Figure 17b, did not remain constant over the different terrains, unlike the open-loop RDS cases, and was seen to generally increase as the surface roughness length increased. This could be in order to attain the higher altitudes while being exposed to lower wind speeds during the ascent. The lift coefficient showed less variation during the lower turns, likely because of the lower variation in wind speeds at lower altitudes.

Compared to the open-loop cases, the closed-loop cases recharge very little to no power during the leeward descent; almost all of the power is regenerated during the upper turn from windward to leeward direction. Figure 18 portrays the location of the applied D_{gen} (yellow) for $P_{net} = 21.6W$

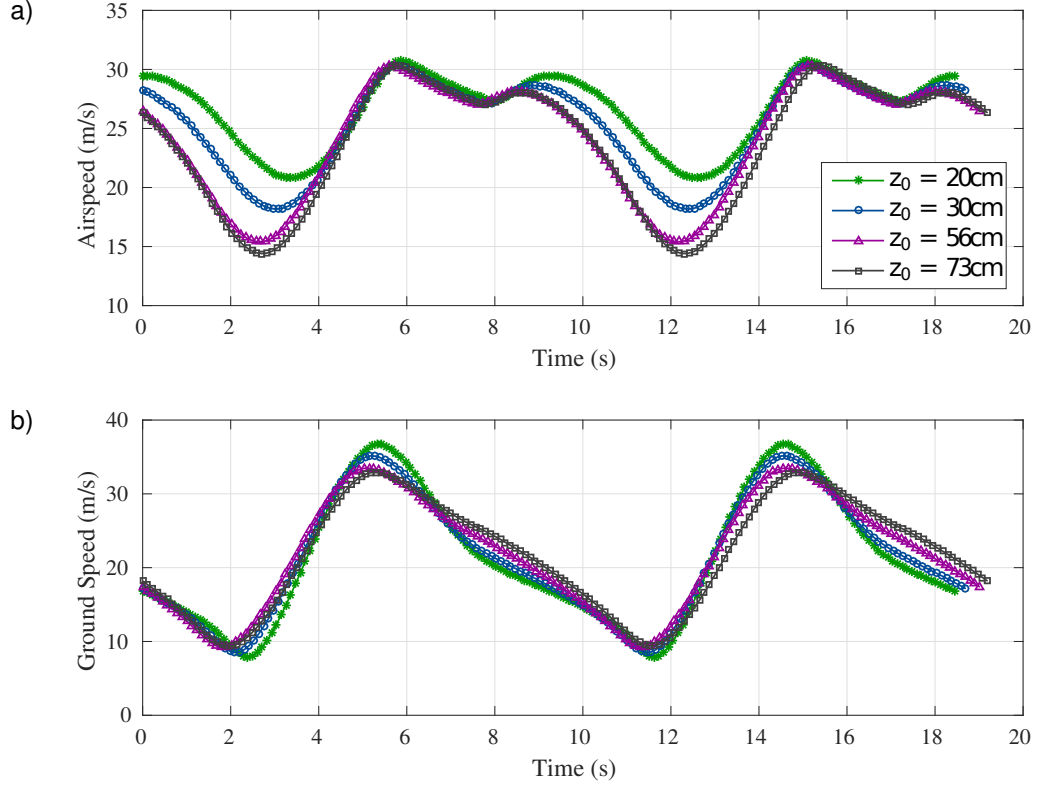


Fig. 16 Time versus: a) Airspeed. b) Ground speed.

over a suburban terrain.

VI. Conclusion

The findings from the RDS trajectory optimization study imply that RDS flight over flat terrains is in fact conceptually feasible. Open-loop RDS over each terrain type investigated was found to be achievable for the full range of P_{net} values tested. The effects of surface roughness on open-loop RDS flight were found to be limited mostly to the variation in wind and ground speeds present during the trajectory. Closed-loop RDS trajectories were also found to be feasible. They were, however, restricted to certain terrain types as a function of P_{net} when it was increased from a value of zero. Only briefly touched on in this paper, a more detailed analysis on the effects of the net power income on RDS trajectories has been accepted for presentation and publication at IMAV2018.

Should the trajectories and power conversion efficiencies detailed in this paper be implementable on an aircraft in flight, then the range and endurance of that aircraft would increase greatly. Using

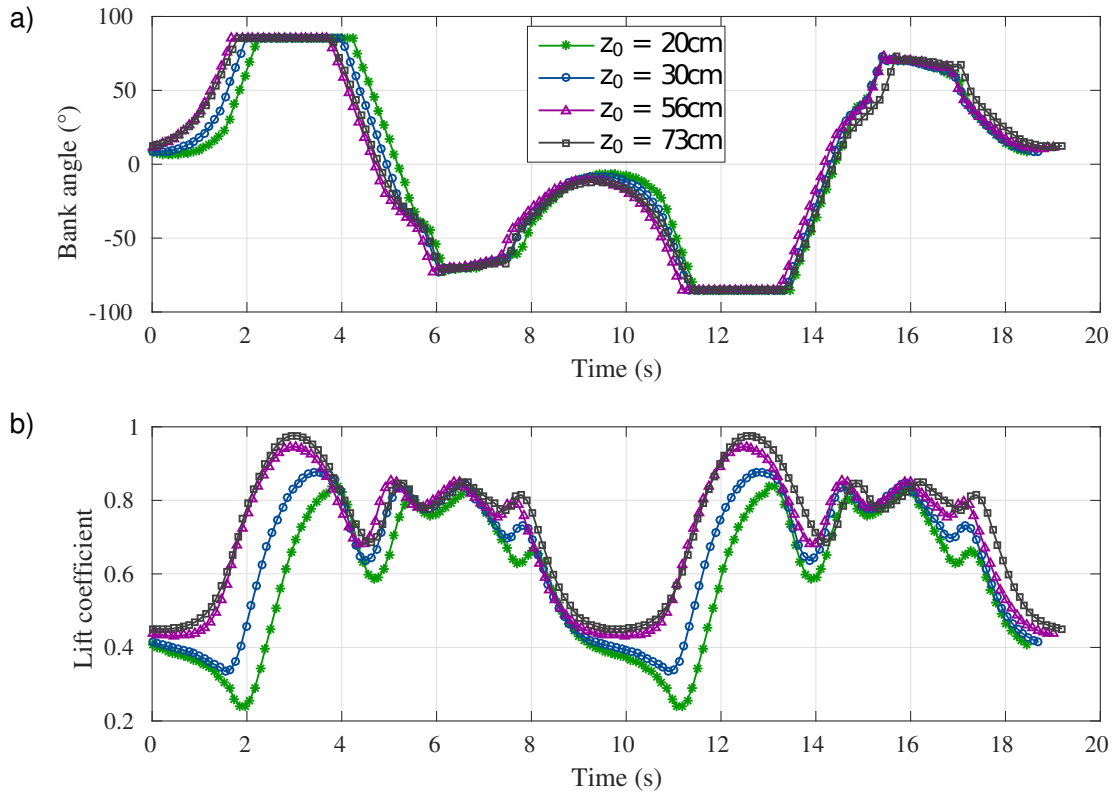


Fig. 17 Time versus: a) Bank angle. b) Lift coefficient.

the DT-18 and the minimum power calculated to keep it in straight and level flight, then for one hour of RDS flight, at least 54 minutes of S&L flight is possible. Moreover, using a combination of open and closed-loop DS, open and closed-loop RDS over flat terrains, and RDS on the leeward side of hills in conjunction with powered flight, then the scope of mission scenarios where endless flight is theoretically achievable expands even more.

These results add to the academic research available on regenerative soaring, expanding the potential scope of RDS, and, thus, taking the concept a step further towards being implemented on an aircraft. However, the simplicity of the constant power conversion efficiency for the wind-milling propeller to battery power chain requires refinement. In order to understand the relationship between power conversion efficiency and the propeller model used, simulations using a BEMT or CFD technique are required, as well as a wind tunnel experimental campaign. The change between propulsive and windmilling modes should be studied, including the 'pinwheeling' state in between the two. Blade optimization should analyze the percentage of time each mode is active to make

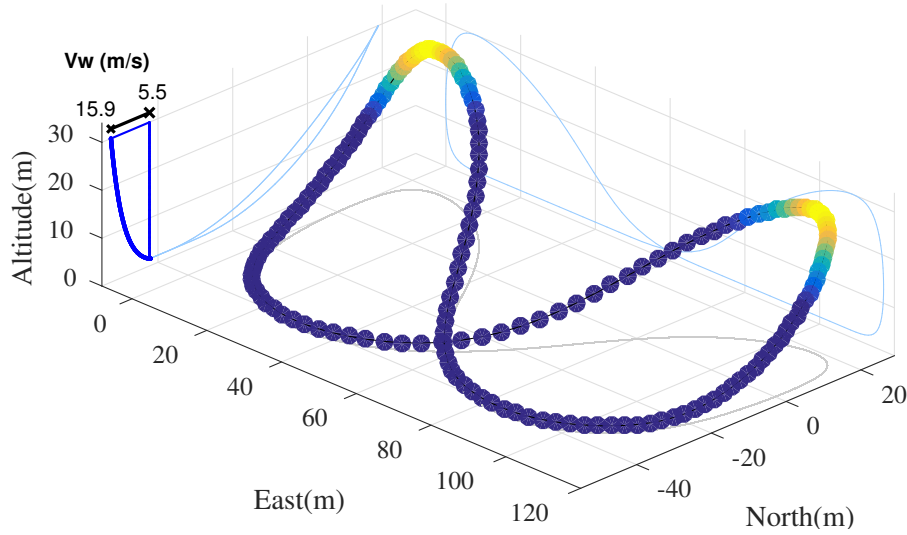


Fig. 18 Regenerative drag force along closed-loop trajectory.

trade offs on the design for high efficiencies in one or both modes. The generator and battery used must also be taken into consideration.

As for the wind model, the effect of obstacles on the wind profiles should be included to refine the model further, as well as investigating the influence of unsteady wind patterns on the trajectories.

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